

Jianyu Zhang

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Research profile

I study baryon spin, fundamental symmetries and heavy-flavour dynamics with BESIII and LHCb. My work combines precision multidimensional angular analysis, CP and electric-dipole-moment measurements, data-driven efficiency calibration and systematic validation. At LHCb, I lead a multibody charm-baryon amplitude analysis.

Appointments

Dec 2025 - present

Postdoctoral Fellow

National Centre for Nuclear Research (NCBJ), Poland

Baryon spin, fundamental symmetries and heavy-flavour research.

Jul 2022 - Dec 2025

Postdoctoral Fellow / Special Research Assistant

University of Chinese Academy of Sciences (UCAS), Beijing, China

BESIII hyperon CP/EDM and interaction studies; LHCb charm amplitude analysis; mentoring and collaboration service.

Education

Sep 2017 - Jun 2022

PhD studies · Particle Physics and Nuclear Physics

Institute of High Energy Physics, Beijing, China

Thesis: *Studies of Hyperon Decays and CP Violation at BESIII.*

Supervisor: Prof. Haibo Li. Degree formally awarded 8 January 2023.

Sep 2013 - Jun 2017

BSc · Physics

Shandong University, Jinan, China

Thesis: *Study of the Process of Lepton Number Violation via $\Xi^- \rightarrow pe^-e^-$.*

Supervisors: Prof. Haibo Li and Prof. Jianbin Jiao.

Selected research contributions

- **Λ electric dipole moment:** contributed to the full angular extraction using entangled $J/\psi \rightarrow \Lambda$ anti- Λ decays, data-driven detector studies and full-fit systematic propagation. The BESIII result is consistent with zero, reaches the 10^{-19} e cm scale and improves the previous upper limit by about three orders of magnitude (Science, 2026).
- **Angular method for hyperon EDM searches:** contributed to the development and experimental implementation of a full angular and spin-correlation strategy for short-lived hyperons (Physical Review D, 2023).
- **Precision CP tests:** played leading roles in hyperon decay-parameter and CP-asymmetry measurements with entangled Λ anti- Λ and Ξ^0 anti- Ξ^0 systems and hyperon-to-neutron decays, including efficiency calibration and systematic validation at the 10^{-3} precision scale.
- **Multidimensional analysis software:** developed a RooFit framework with unbinned fits, pseudo-data generation, CPU multithreading and precomputed detector-acceptance integrals; adopted in nine additional BESIII analyses.
- **Hyperon interactions:** co-developed collider strategies using detector material to study secondary hyperon scattering and antihyperon-proton annihilation.
- **Charm amplitude analysis:** leading the ongoing $\Lambda_c^+ \rightarrow \Lambda \pi^+ \pi^+ \pi^-$ analysis at LHCb, with experience in multidimensional amplitude models, large-scale simulation and collaboration validation.

Experimental collaborations & service

- **BESIII, 2017-present:** hyperon spin, decay parameters, CP/EDM measurements and hyperon-nucleon interactions. Member of internal review committees for seven collaboration papers; regular data-acquisition shifts and detector/systematic studies.
- **LHCb, 2022-present:** multibody charm amplitude analysis. Charm Working Group MC liaison during 2024-2026, supporting simulation production, validation and cross-analysis coordination.

Analysis expertise

Multidimensional angular and helicity-amplitude fits; spin-sensitive observables; polarisation and sequential-decay analysis; CP/EDM null tests; data-driven efficiency calibration; detector acceptance and systematic propagation; RooFit likelihoods and pseudo-data generation; CPU multithreading; TF-PWA / TensorFlow and GPU workflows; simulation production and validation.

Selected publications

01. Measurement of the Λ Electric Dipole Moment with an Entangled Baryon-Antibaryon System

BESIII Collaboration · Science 393, 1027-1030 (2026)

arXiv:2506.19180

02. Probing hyperon electric dipole moments with a full angular analysis

J. Fu, H. B. Li, J.-P. Wang, F.-S. Yu, J. Zhang · Phys. Rev. D 108, L091301 (2023)

arXiv:2307.04364

03. Precise Measurements of Decay Parameters and CP Asymmetry with Entangled Λ -anti- Λ Pairs

BESIII Collaboration · Phys. Rev. Lett. 129, 131801 (2022)

arXiv:2204.11058

04. Tests of CP symmetry in entangled Ξ^0 -anti- Ξ^0 pairs

BESIII Collaboration · Phys. Rev. D 108, L031106 (2023)

arXiv:2305.09218

05. Test of CP Symmetry in Hyperon to Neutron Decays

BESIII Collaboration · Phys. Rev. Lett. 131, 191802 (2023)

arXiv:2304.14655

06. Observation of anti- $\Lambda p \rightarrow K^+\pi^+\pi^-\pi^0$ and anti- $\Lambda p \rightarrow K^+\pi^+\pi^-2\pi^0$

BESIII Collaboration · Phys. Rev. Lett. 136, 171904 (2026)

arXiv:2602.01282

07. First Study of Reaction $\Xi^0 n \rightarrow \Xi^- p$ Using Ξ^0 -Nucleus Scattering at an Electron-Positron Collider

BESIII Collaboration · Phys. Rev. Lett. 130, 251902 (2023)

arXiv:2304.13921

08. First measurement of ΛN inelastic scattering with Λ from $e^+e^- \rightarrow \Lambda$ anti- Λ

BESIII Collaboration · Phys. Rev. C 109, L052201 (2024)

arXiv:2310.00720

09. Prospects to study hyperon-nucleon interactions at BESIII

J. Dai, H. B. Li, H. Miao, J. Zhang · Chin. Phys. C 48, 073003 (2024)

arXiv:2209.12601

10. Searches for New Physics Beyond the Standard Model in Hyperon Sector

J. Zhang, J. Fu, H. B. Li · Chin. Phys. Lett. 43, 060201 (2026)

arXiv:2602.22119

Selected presentations

- **30 June 2026:** Hyperon Physics at BESIII. MESON 2026, Kraków, Poland. Invited plenary.
- **12 November 2025:** Hyperon Physics at BESIII. Baryons 2025, Jeju, Korea. Invited plenary (remote).
- **2 July 2025:** Highlight on CPV test of hyperon at BESIII. High-Energy Physics International Conference in Quantum Chromodynamics, Montpellier, France.
- **21 November 2024:** Searching for Hyperon Electric Dipole Moments at STCF. International Workshop on Future Tau Charm Facilities, Guangzhou, China.
- **16 October 2023:** Probe hyperon electric dipole moments at e^+e^- colliders. 16th International Conference on Meson-Nucleon Physics and the Structure of the Nucleon, Mainz, Germany.

Awards & recognition

- **2024:** First Prize, 13th “Chenguang Cup” Young Outstanding Paper Award, High Energy Physics Division of the Chinese Physical Society.
- **2022:** BESIII PhD Thesis Award; Outstanding PhD Thesis, IHEP.
- **2022-2025:** Outstanding Individual in Annual Assessment, UCAS.
- **2022:** Best Poster, 13th International Workshop on e^+e^- Collisions from Phi to Psi, Shanghai.

Teaching & mentoring

- Mentored research projects involving two Master’s and two PhD students during the UCAS postdoctoral period. Guided an undergraduate thesis that received a university-level Outstanding Undergraduate Thesis Award.
- Teaching Assistant, Atomic Physics, UCAS (2019, 2020, 2022); Outstanding Teaching Assistant recognition.
- Instructor, UCAS Summer Science and Innovation Practice in Particle Physics Experiments (2024), covering statistical methods and analysis tools.